```
for i in range(len(flowers)):
    if s==flowers[i]:
      print(s, " found at index number ",i)
      break
else:
print("Sorry' Not Found")
```

Tuples in Python

In the previous chapter, you have learned about list. Similar to list, a tuple is a sequence of values enclosed in parentheses and its indices start with 0. Unlike list, tuple cannot be changed which means it is immutable. In this chapter, we will discuss how to create, access and perform operations on tuples.

Creation of Tuples

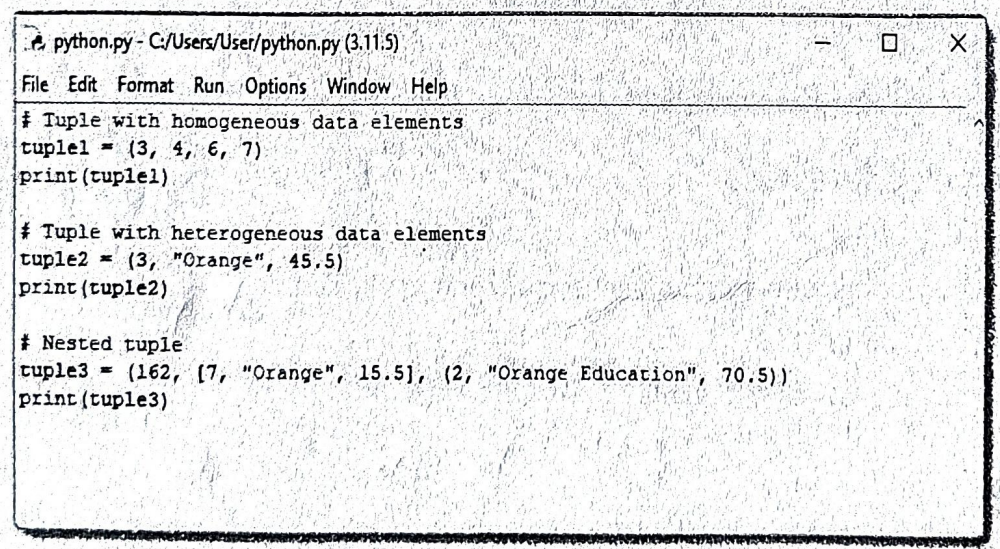
To create a tuple, write different values and separate them with comma. You can put the comma-separated values between the parentheses '()'. Parentheses (round bracket) indicate the start and end of the tuple.

Tuple contains any number of elements of heterogeneous type (integer, float, string, list, etc).

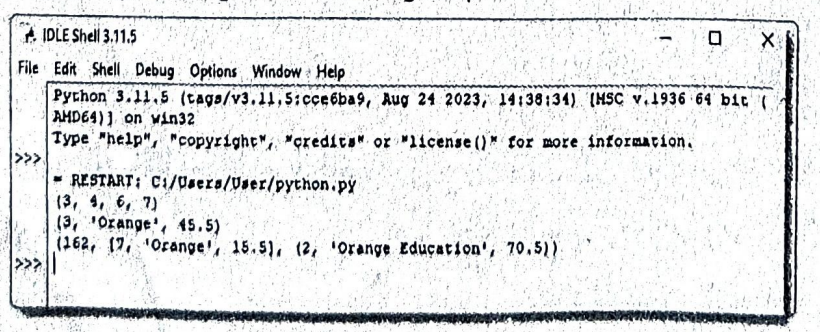
Example of tuple:

```
tup1=('sst', 'chemistry', 2010, 2022); #tuple with dissimilar items
tup2=(1, 2, 3, 4, 5 );                 #tuple with similar items
tup3=(1, 2.5, 3.6, 4, 5 );             #t 50, tup1 is treated to be of type int.
```

Program 1: To create a tuple.



On running the above program, you will get the following output:



ntax to create a tuple:

```
= (value1, value2, value3, ...)
```

To create an empty tuple: `tup= tuple()`

To create a tuple with single element: `tup= (1,)`

To create long tuple: `tup = (0, 1, 2, 3, 4 , 5, 6, 7, 8, 9 10, 11, 12, 13, 14, 15)`

To create nested tuple: `tup = (1,2, (3,4))`

To create tuple by using built-in tuple type object: `tup = tuple(<sequence>)`

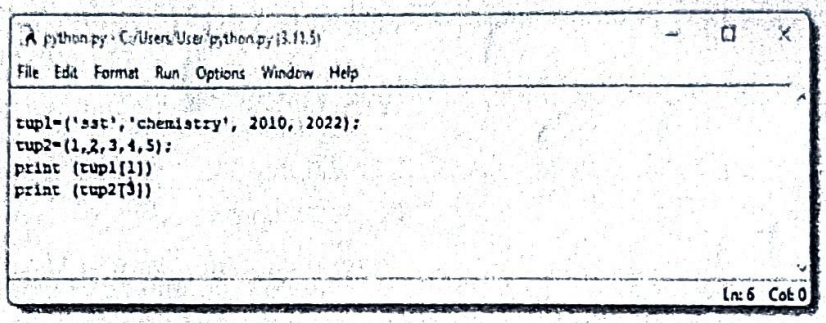
here, sequence is an object which includes strings, lists and tuples.

```
tup= tuple('orange')
tup
', 'r', 'a', 'n', 'g','e')  #tuple tup is created from other sequence- a string
                            'orange'.
```

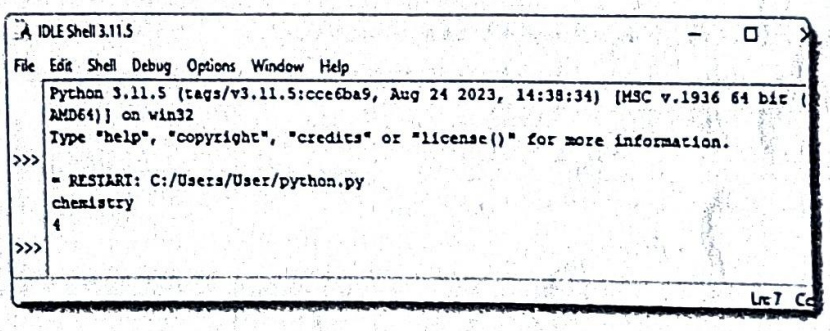
ccessing Tuples

access values in a tuple, use the square brackets along with the index number for reference. So, you can access tuple element just like you access a list's or string's element.

ogram 2: To access a value in tuple.



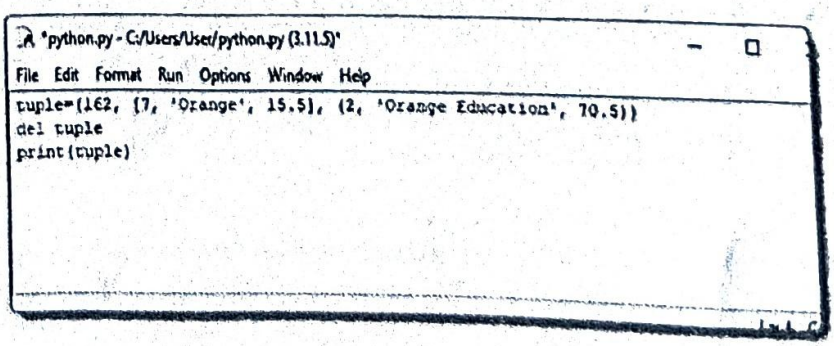
running the above program, you will get the following output:



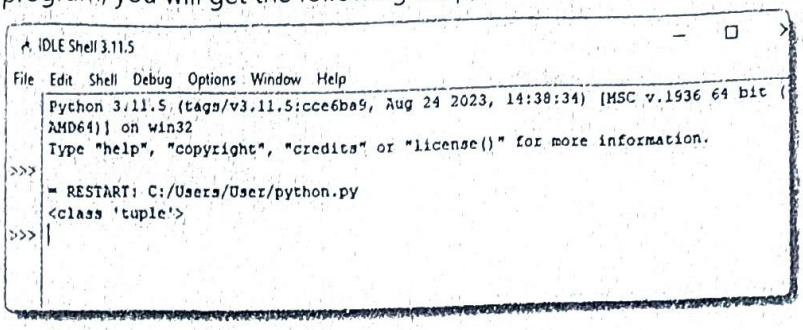
eleting a Tuple

ements of a tuple cannot deleted or removed, but we can delete the entire tuple by using the **del** keyword.

ogram 3: To delete a tuple.



On running the above program, you will get the following output:

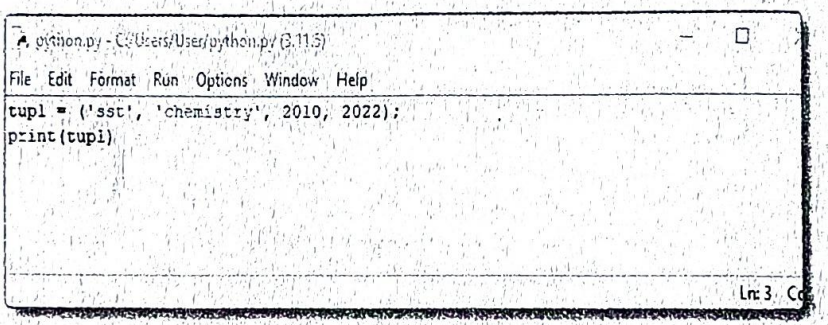


Difference between List and Tuple

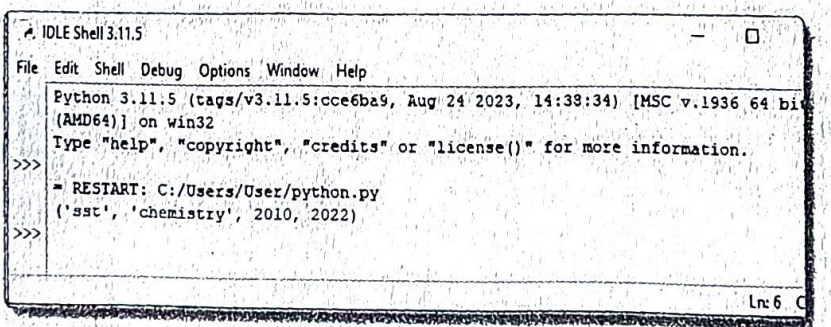
List	Tuple
A list is a sequence of multiple values in an ordered sequence.	A tuple is a collection of objects which is ordered.
List is enclosed in square brackets [].	Tuples are enclosed in parenthesis ().
Elements of the list can be changed.	Elements of the tuple cannot be changed.
List has the variable length.	Tuple has the fixed length.
List is mutable.	Tuple is immutable.
List can hold only homogeneous values.	Tuple can hold both homogeneous and heterogeneous values.

Some More Programs

Program 4: To print the entire tuple.



On running the above program, you will get the following output:



ogram 5: To use tuple method count and index.

```
File Edit Format Run Options Window Help
tup=(2,5,7,8,2,1)
print(tup.count(2))
print(tup.index(7))
```

nverting list to tuple and vice-versa

ning the above program, you will get the following output:

```
IDLE Shell 3.11.5
File Edit Shell Debug Options Window Help
Python 3.11.5 (tags/v3.11.5:cce6ba9, Aug 24 2023, 14:38:34) [MSC v.1936 64 bit (
AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/Users/User/python.py
2
2
>>>
Ln: 7 Col: 0
```

ogram 6: **A tuple "tup" stores (2,5,7,8,2,1) by mistake its last element is typed wrong. Write a program correct its last element as 11.**

```
File Edit Format Run Options Window Help
tup = (2, 5, 7, 8, 2, 1)
tupl = list(tup)
tupl[-1] = 11
tup = tuple(tupl)
print("Modified tuple:", tup)
Ln: 8 Col: 0
```

n running the above program, you will get the following output:

```
IDLE Shell 3.11.5
File Edit Shell Debug Options Window Help
Python 3.11.5 (tags/v3.11.5:cce6ba9, Aug 24 2023, 14:38:34) [MSC v.1936 64 bit (
AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/Users/User/python.py
Modified tuple: (2, 5, 7, 8, 2, 11)
>>>
Ln: 6 Col: 0
```

Converting List to Tuple and Vice-versa

- list can be converted into a tuple by using a built-in function tuple().

or example:

```
L=[1,2,3,4]
type(L)
ass 'list'>
```

```
>>> T=tuple(L)
>>> type(T)
<class 'tuple'>
```

A tuple can be converted into a list by using a built-in function list()

For example:

```
>>> L=[1,2,3,4]
>>> type(L)
<class 'list'>
>>> T=tuple(L)
>>> type(T)
<class 'tuple'>
>>> L2=list(T)
>>>L2
[1, 2, 3, 4]
>>> type(L2)
<class 'list'>
```